WildTrace: Learning Sub-Millimetre Pen Path Following Using Kinematic IK Labels

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***Abstract*—We investigate whether a learned policy can track pen paths more accurately than the inverse-kinematics (IK) controller used to generate its training labels. WildTrace converts photographs into ordered pen strokes, producing 2,874 drawings across six animal classes. An ACT-based policy uses joint state and a short window of the upcoming path to predict joint increments, trained solely on kinematic IK labels with injected noise. In a calibrated Isaac Sim twin of a 7-DoF xArm 7, the policy completes all 48 drawings excluded from training, compared with 45/48 for Lula IK. It reduces mean path RMSE from 0.277 to 0.192 mm, with lower error on all 45 paired drawings, and halves joint jerk. Removing the settle gate through a streaming IK baseline improves RMSE by only 0.010 mm. A fine-tuned SmolVLA achieves 0.180 mm in a single training seed, but with higher jerk and query latency. On the physical arm, the twin-trained policy completes 30 animal drawings with a mean RMSE of 0.532 mm; 14 require manual nudges to resolve stalls. We analyse the approximately 2.7-fold difference from the twin. In nine paired real-arm drawings, ACT draws 4.9 times faster than IK (91 against 446 s) but leaves more gaps in the ink (2 of 9 sheets complete against 8 of 9, judged from photographs).**

***Index Terms*—robot drawing, imitation learning, action chunking, vision-language-action models, digital twin, sim-to-real**

I. Introduction

Robot drawing is a compact testbed for precise contact-rich path following: the pen must stay on a sheet a fraction of a millimetre away, follow tight curves without overshoot, and lift and land cleanly between strokes. Every error is visible on paper. Classic pipelines solve inverse kinematics (IK) per waypoint and wait for the arm to settle [1], [2], which is reliable but slow and leaves a visible tremor. Learned visuomotor policies such as ACT [3], diffusion policy [4] and vision-language-action (VLA) models such as SmolVLA [5] promise smoother motion, but are usually evaluated by task success rather than by sub-millimetre path accuracy on open-ended content.

WildTrace closes the loop from an arbitrary photograph to ink on paper (Fig. 1) and uses it to compare controllers on the same drawings under identical physics. Our contributions are:

1) A photo-to-trajectory pipeline with a generate-judge-retry loop, which we use to build 2,874 validated drawing trajectories and a 48-drawing held-out test set (Sec. III).
2) A path-conditioned ACT-based policy trained only on kinematic IK labels with noise injection and a capped progress projection, which beats IK in a calibrated 1 g twin on accuracy, completion and smoothness, maintains comparable error across the tested paper placements, and keeps its lead over a streaming IK baseline that shares its progress rule and pace (Sec. IV, V).
3) A like-for-like comparison with a fine-tuned SmolVLA on the same data: re-planned every tick it matches and slightly exceeds ACT on the test set, at higher jerk and latency, and a negative result on camera-conditioned gap repair.
4) A hybrid supervisor that repairs skipped ink using the run's own bookkeeping, and a physical-arm evaluation on 30 animal drawings and three portraits that quantifies the sim-to-real gap, with a paired comparison against IK on nine drawings that shows where the learned policy still falls short (Sec. V).

II. Related Work

Robot drawing and painting. Portrait-drawing robots date back to humanoid systems that extract contours and replay them with IK [1] and to Paul, which emulates an artist's sketching process [2]. RoboCoDraw turns a face into a GAN avatar and orders its strokes with a genetic algorithm, drawing in 43 s on a UR5 [9]; AiSketcher draws synthesised portraits with a 3-DoF arm in about two minutes [10]; Nasrat et al. draw portraits in 80 s with line width set by pen height [11]; Piktor-O-Bot adds force-controlled shading on a UR5 and shapes speed by path curvature [12], [13]; and TSP-Bot draws single-path stipple art with TRAC-IK [14]. These systems execute planned paths and report drawing time, style or viewer preference rather than tracking accuracy. Learned stroke models [15], [16] plan in image space, and FRIDA [17] and Spline-FRIDA [18] learn stroke appearance to shrink the sim-to-real gap of the painted image. Closest to our control question, Lee et al. learn motor control with hierarchical reinforcement learning instead of IK, but need pen-tip feedback to correct the learned controller on a real UR5e [19]. We take the strokes as given and measure how precisely learned policies follow them against IK, at sub-millimetre resolution.

Imitation learning for fine control. ACT [3] predicts chunks of future actions with a transformer; diffusion policy [4] models multimodal action distributions. DART [20] injects noise during demonstration to cover off-path states and DAgger [21] aggregates on-policy corrections. We use DART-



Fig. 1. WildTrace data pipeline on one Open Images [6] photo. Bronze: fetch the image, mask and license, then curate and label (BioCLIP [7]). Silver: isolate the subject and redraw it as an outline with FLUX [8], judged and retried. Gold: the accepted outline becomes a few ordered pen strokes on a normalised canvas (numbered markers show stroke starts).

style noise with a synthetic IK teacher, and DAgger rounds for camera policies.

Precision tracking and residual learning. Iterative learning control improves commands for a repeated trajectory using errors from previous executions [22]. Model predictive contouring control balances contour error with progress along the path [23]. Both provide strong conventional alternatives for precision drawing. Residual policy learning augments a conventional controller with a learned correction [24], [25]. Our policy replaces the controller and learns corrective actions from noise-injected kinematic labels. Direct comparisons with these alternatives remain future work.

Vision-language-action models. SmolVLA [5] reuses a pretrained vision-language backbone for control, with a flow-matching action head [26]. We fine-tune SmolVLA on the same data as ACT to test whether a pretrained generalist helps a precision task.

Sim-to-real. Domain randomisation [27] widens the training distribution to cover the real system; we instead calibrate the twin to the lab arm and randomise only paper placement.

III. Data Pipeline: Photo to Pen Strokes

The pipeline is a local-first medallion design with one script per stage and append-only manifests, so a stage reruns only for rows whose configuration or input hash changed.

Bronze. Images, segmentation masks and licenses are fetched from Open Images V7 for six classes (bird, cat, dog, fish, frog, horse), de-duplicated, filtered for quality and viewing angle, and labelled with BioCLIP.

Silver. The subject is cropped by its mask and redrawn as a black-on-white outline by FLUX.1-schnell. A LangGraph [28] loop validates each outline with an OpenCV prescreen (foreground ratio, component count, contour size, fragmentation) followed by a VLM judge, and retries with a new seed on failure.

Gold. The best outline per pose is skeletonised and vectorised into a few ordered strokes, resampled and normalised to a unit canvas. The result is 2,874 validated trajectories. A stratified 48-drawing gold test set (9 bird, 12 cat, 13 dog, 4 fish, 5 frog, 5 horse; mean 4.6 strokes, range 2 to 8, mean 370 points) is never used for training, but was used during development to compare design variants (Table III). Faces follow a separate CPU path: MediaPipe [29] face landmarks and hair/body segmentation are traced into a pen-sketch portrait. Fig. 2 shows five portraits from this path next to their source photographs.

TABLE I
Outline generator and judge study (20 subjects, referee Qwen3-VL-235B).

Generator	Referee pass	BioCLIP top-1
FLUX.1-schnell	**20/20**	**75%**
OmniGen2	6/20	30%
Judge	**Agreement**	**Cohen's κ**
Qwen2.5-VL-7B (local)	passes 9/9 blanks	0.00
Qwen3-VL-8B	**92.5%**	**0.83**

Choosing the generator and the judge. Table I summarises a 20-subject study against a strong referee (Qwen3-VL-235B [30]). FLUX produced acceptable outlines for all 20 subjects against 6/20 for OmniGen2 [31] (paired sign test $p = 10^{-4}$), whose failures were mostly blank pages. The local Qwen2.5-VL-7B [32] judge originally in production agreed with the referee no better than chance and passed every blank page; we therefore switched to Qwen3-VL-8B, which agrees with the referee on 92.5% of outlines.

Why open-weight models. Every generator and judge runs locally on a single 16 GB laptop GPU, and we did not evaluate proprietary frontier models in either role. This keeps the thousands of generate-judge calls at zero marginal cost, keeps uploaded faces off third-party servers, and gives reproducible verdicts from pinned weights; an 8B open judge already agrees with a 235B open referee at $\kappa = 0.83$. The cost is that we cannot report how a frontier judge would score; the referee is itself an open model.

IV. Robot, Twin and Controllers

Hardware and twin. A UFACTORY xArm 7 holds a pen in its parallel gripper over a US Letter sheet. The Isaac Sim [33] twin (Fig. 3a) uses the arm's calibrated URDF, the measured paper pose and pen touch height, Earth gravity, and a stiff PhysX position servo (stiffness 7.5×10^5, damping 2050) standing in for the vendor joint controller. Two RealSense D435i cameras (wrist and overhead) were located by depth plane fitting and planar PnP on the sheet corners, whose

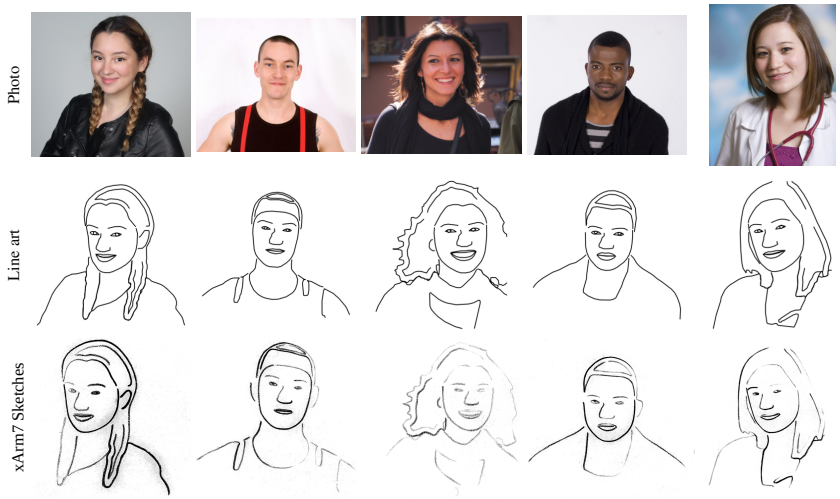


Fig. 2. Face portraits: Open Images source photographs (top), the pen-sketch line art traced from MediaPipe landmarks and segmentation (middle) and the xArm 7 sketches (bottom), 15 to 19 strokes each. Photographs CC BY 2.0 via Flickr, by Byron Gonzalez, MangakaMaiden Photography, Luca Boldrini, JJP Bolger Photography and Walt Stoneburner (left to right).



(a) Lab-arm twin (b) Physical xArm 7 (c) Real portrait

Fig. 3. (a) Isaac Sim twin built from the lab arm's calibrated URDF, gripper and measured paper placement, at 1 g. (b) The physical xArm 7 drawing a WildTrace bird. (c) A face portrait drawn by ACT on the physical arm with a marker (portrait A in Table VII).

heights above the paper agree to within 1.6%. Each drawing runs a fixed phase machine: approach, lower, draw, lift for every stroke, with short pen-up hops between nearby strokes.

Metrics. Let $\{p_i\}_{i=1}^{N}$ be the planned pen-down samples and $\{e_j\}$ the executed pen-down pen-tip positions, both in 3-D. The nearest-path error of a planned sample is

$$d_i = \min_j \|p_i - e_j\|_2, \tag{1}$$

and we report the path RMSE $\left(\frac{1}{N}\sum_i d_i^2\right)^{1/2}$, the 95th percentile and the maximum of d_i. Because (1) only penalises missing a planned point, we also report the reverse direction, from each executed sample to the plan, which catches stray ink. Joint jerk is the RMS over pen-down samples of $\|\Delta^3 q\|/\Delta t^3$, the third difference of joint positions within a stroke. Corner RMSE is the tracking error at waypoints flagged as corners, and heading error compares the direction of plan and pen over a 4 mm chord.

IK baseline. NVIDIA Lula IK solves joint targets for each waypoint of the resampled path (at most 5 mm apart, densified at corners); joint commands are slew-limited, and the run advances only when the measured tip is within tolerance and its velocity has settled. A waypoint timeout fails the drawing safely. This is the controller the system shipped with and the teacher that labels the policy data.

Streaming IK baseline. To separate learning from the settle gate, a second IK baseline uses exactly ACT's progress rule (the first waypoint ahead of the pen's projection, Sec. IV) with no settle gate, and solves Lula IK every tick for a carrot a fixed arc length L ahead of the pen (pure pursuit). The carrot advances along the path no faster than ACT's median pen speed, 12.4 mm s^{-1}, so both controllers draw at the same pace; we also run it at plain IK's median pen speed, 6.9 mm s^{-1}. On six drawings outside the test set, L between 0.5 and 2 mm made no difference because the speed limit binds, so we use $L = 0.5$ mm. Without the speed limit, pure pursuit raced through a drawing in a sixth of the time at 0.69 mm RMSE.

ACT-based path follower. At control tick t the policy observes the robot state $s_t = [\,q_t,\ \dot{q}_t,\ u_{t-1}\,] \in \mathbb{R}^{21}$ (joint positions, velocities and previous joint command) and a path context

$$c_t = \left[\,x_t,\ g_t - x_t,\ \{w_{t,k} - x_t\}_{k=1}^{10},\ \pi_t\,\right] \in \mathbb{R}^{37}, \tag{2}$$

where x_t is the pen tip, g_t the current target, $w_{t,k}$ ten points spaced at equal arc length along the next 2 cm of path, and π_t the pen state. It predicts a chunk of $H = 10$ joint *increments* $\hat{\delta}_{t:t+H-1}$, trained with an L1 loss against the teacher's increments, and only the first is executed:

$$u_t = u_{t-1} + \alpha\,\hat{\delta}_t, \qquad \alpha = 0.5. \tag{3}$$

Absolute joint targets were mm-imprecise even when overfitting one drawing, so the policy predicts increments. There is no image input. We refer to the policy as ACT-based because it omits ACT's CVAE and temporal ensembling and uses robot state and path information instead of images. For brevity, we use ACT below. We use ACT [3] via LeRobot [34] with $d = 256$, 4 encoder and 1 decoder layers, no CVAE, no dropout, learning rate 3×10^{-4}, batch 256 and 40k steps. On a

single-drawing overfit gate these choices cut first-action error from 33-45% to 11% of the label, a 10-step chunk fit 2.5× better at p95 than a 30-step chunk, and executing only the first action of each chunk with re-planning every tick tracked better than temporal ensembling. At run time the policy is queried at 60 Hz and half of each predicted step is applied.

Labels without physics. Labels come from Lula IK walked through the same phase machine with no physics simulation (about 20k solves/s), three episodes per training drawing. Two of the three add DART-style command noise [20] and a measured-versus-commanded joint offset, so the policy sees sagged, off-path states with the corrective label. Path progress is the projection of the tip onto the path, capped at 1.5 mm ahead; without the cap progress jumped across narrow hairpins (8.1 mm worst-point error on one frog toe) and spike tips were rounded. The lab-arm model is trained with ±15 mm paper-height jitter. Fine-tuning on 1 g Isaac IK rollouts made tracking worse (0.74 to 0.91 mm on four drawings), because those labels carry IK's own sag.

SmolVLA. We fine-tune SmolVLA [5] on the same dataset (19.8M frames), feeding the full 58-value state through a direct-state MLP, 8 VLM layers, 256 px rendered path images and 20-action chunks (51M of 323M parameters trained). It is served with zero initial noise (better than averaging samples offline); with 3 flow-matching steps, a captured CUDA graph cuts sampling from about 225 ms to 30 ms per chunk.

Hybrid supervisor. A rule-based supervisor watches the run's own ink bookkeeping rather than the policy. If the drawing target has not advanced for 1.5 s, IK drives the next 5 mm and hands back. If at least 1.5 mm of path behind the pen has no ink within 1 mm (the labeller's coverage rule), IK goes back past the gap and forward again while the run's progress is held, then the policy resumes. Both rules need no camera, so they also operate on the physical arm. However, the gap rule detects only gaps represented in the ink bookkeeping and cannot identify physical ink failures such as a dry pen.

V. Results

All simulated runs use the lab-arm twin at 1 g. The main comparison (Table II) runs all four controllers on all 48 gold drawings with the same code, scene and batch runner; per-run metrics and plots are kept with the code.

ACT against IK. ACT finishes every drawing; IK fails three on waypoint timeouts. On the 45 drawings both finish, ACT has lower path RMSE on all 45: the paired reduction is 0.085 mm (95% CI 0.079 to 0.091), or 31%, with two-sided Wilcoxon signed-rank $p = 5.7 \times 10^{-14}$. The advantage holds in every class (ACT 0.183 to 0.198 mm, IK 0.266 to 0.295 mm). It is not bought with stray ink: the reverse, pen-to-plan error is also lower (0.374 against 0.397 mm). Over pooled pen-down samples the median error falls from 0.256 to 0.141 mm (0.120 mm for SmolVLA). ACT halves joint jerk and removes the per-waypoint tremor visible in Fig. 4. IK keeps a lower heading error (median 0.51° against 1.36°), since it settles on each waypoint: its pen is settling on 51% of pen-down samples against 18% for ACT, and moves at a median 6.9 against 12.4 mm s^{-1}. Total draw time is similar (77 against 73 s simulated) because approach, lower and lift moves are shared. The policy server answers in 9.1 ms median, inside the 16.7 ms tick. Retraining ACT with two more seeds gives 0.201 and 0.218 mm on the same 48 drawings (three seeds: 0.204 ± 0.013 mm), all well below both IK variants.

TABLE II

Gold test set, 48 held-out drawings, lab-arm twin at 1 g, same code and runner for all four controllers. Means over finished drawings; 95% bootstrap intervals over drawings in brackets. S-IK is the streaming IK baseline at ACT's pace.

Metric	Lula IK	S-IK	ACT	SmolVLA
Finished	45/48	**48/48**	**48/48**	**48/48**
Path RMSE (mm)	0.277	0.267	0.192	**0.180**
95% CI	.271-.283	.262-.272	.190-.194	.176-.184
Path p95 (mm)	0.398	0.373	0.335	**0.327**
Path max (mm)	0.778	0.953	**0.776**	0.871
Reverse RMSE (mm)	0.397	0.401	0.374	**0.356**
Corner RMSE (mm)	0.366	0.608	**0.325**	0.330
Heading, median (°)	0.51	**0.41**	1.36	1.10
Jerk RMS (rad s^{-3})	757	745	**390**	521
Samples settling (%)	51	48	18	**17**
Median speed (mm s^{-1})	6.9	7.8	12.4	**14.1**
Sim. draw time (s)	76.8	101.0	73.4	**71.3**
Latency p50 (ms)	-	-	**9.1**	17.7

TABLE III

How the ACT path follower got there (48 gold drawings, 1 g, means over finished drawings, earlier code). Rows above the rule use the default sim scene, rows below the stiffened lab-arm twin.

Controller	Fin.	RMSE	Max	Jerk
Lula IK	39/48	0.698	0.985	806
ACT, 30 Hz, full step held 2 ticks	48/48	0.366	2.599	1018
+ 60 Hz queries, half step (Eq. 3)	48/48	0.325	2.360	272
+ progress capped at 5 mm (hairpins)	48/48	**0.239**	1.472	**255**
+ cap at 1.5 mm (spike tips)	48/48	0.256	**0.885**	285
Lula IK, lab-arm twin	44/48	0.283	**0.775**	752
ACT, lab-arm twin, height jitter	**48/48**	**0.195**	0.812	**390**

Is it the learning or the settle gate? Streaming IK uses ACT's progress rule with no settle gate. It finishes all 48 drawings (IK's timeouts came from the gate), but its path RMSE improves on IK by only 0.010 mm (0.267 mm, lower on 37 of 45), its corner error grows from 0.366 to 0.608 mm because pure pursuit cuts corners, and its jerk stays at IK's level. ACT is lower than streaming IK on all 48 drawings (paired reduction 0.075 mm, CI 0.069 to 0.080). Slowing streaming IK to IK's own pace (carrot limit 6.9 mm s^{-1}) does not help: 0.266 mm RMSE, jerk down to 524 but corner error up to 0.973 mm, and 181 s per drawing, still above ACT on all 48 drawings. Removing the gate is not what makes ACT accurate when compared with these two IK controllers.

What mattered. Table III traces the design. Querying at 60 Hz with half steps cut jerk 3.7×; capping the progress projection fixed hairpin jumps (mean worst point 2.36 to 1.47 mm), and a tighter cap fixed spike tips (0.885 mm) at a small RMSE cost. Moving to the stiffened lab-arm twin,

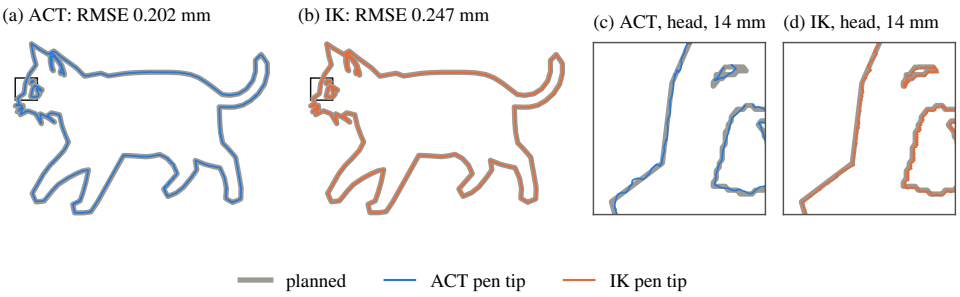


Fig. 4. Planned path (grey) against the executed pen tip for the same gold cat, ACT (blue) and IK (orange), top-down. (c, d) A fixed 14 mm window on the head: IK's stop-and-settle at each waypoint leaves a tremor of small hooks along the stroke; ACT stays smooth.

TABLE IV
PAPER PLACEMENT ROBUSTNESS, 16 GOLD DRAWINGS (FINISHED, MEAN PATH RMSE IN MM).

Placement	Lula	IK	ACT	
Nominal	15/16	0.278	16/16	0.192
12 mm lower	13/16	0.284	16/16	0.189
12 mm higher	15/16	0.271	16/16	0.193
Shifted $(+15, -15)$ mm	16/16	0.300	16/16	0.190

with paper-height jitter in training, gives the final policy. Each variant beats IK on path RMSE in its own scene.

Paper placement. Table IV moves the sheet on a 16-drawing subset. ACT, trained with height jitter, finishes all 64 runs within 0.189 to 0.193 mm. IK drops to 13/16 with the paper 12 mm lower and degrades to 0.300 mm under a planar shift.

SmolVLA against ACT. On the full gold set with the batch runner (Table II), SmolVLA stage 2, re-planned every tick with zero noise, three flow steps and a CUDA graph, is slightly more accurate than ACT: 0.180 against 0.192 mm (SmolVLA uses one training seed; ACT has an across-seed standard deviation of 0.013 mm), lower on 36 of 48 drawings (paired reduction 0.012 mm, CI 0.008 to 0.017, $p = 5\times10^{-6}$), and best in every class. It pays with a third more joint jerk (521 against 390), a higher worst point (0.871 against 0.776 mm), and twice the query latency (17.7 against 9.1 ms). In the interactive runner used for the studies below the order was reversed (0.230 against 0.206 mm on six animals, 10 flow steps; 0.252 against 0.206 mm on one cat with the three-step CUDA-graph server). The runners share the policy servers but have separate control loops and we have not isolated which difference flips the order, so we compare only within a runner. These studies use one gold drawing per class and six face sketches unseen by both policies. Executing three actions per chunk gives 0.327 mm; re-planning every tick gives 0.230 mm, within 12% of ACT, so most of the gap was execution, not prediction. On the six faces SmolVLA reaches 0.288 mm against 0.252 mm for ACT (14% behind), and both generalise from animals to faces at about +22 to +25% RMSE, although faces were never in either policy's training data.

Camera-conditioned SmolVLA (negative result). Replacing the rendered path with synthetic overhead and tip-crop camera views improved offline error (0.306) but not closed-loop tracking, and the policy ignored the page: blanking it changed first-action error by less than 0.001. Gap-labelled data, two DAgger rounds [21], retrace weighting and a redraw mode taught the policy to reverse direction at a gap (2% of going-back actions pointed the wrong way, against 38% for ACT), but not to land back on the line: the camera policies abandoned 16 of 17 self-directed redraws.

Hybrid repair. With the pen made to skip ink on purpose (3 to 8 mm skips every 60 mm of inked travel on average), the supervisor made 87 redraws for 105 skips under ACT across 12 drawings, gave up none, and left at most 0.49% of any path uninked, at a cost of at most 0.011 mm RMSE and 0.4 to 2.2 s per drawing. Without skips it never intervened.

Speed. Scaling ACT's applied step α of Eq. (3) from 0.5 to 0.75 keeps RMSE within 6% (0.206 to 0.219 mm) and cuts simulated time by 24%; 2.0 overshoots and trips the 5 mm safety stop, consistent with the curvature-speed trade-off of Piktor-O-Bot [13]. SmolVLA has no faster setting within 10%.

Physical arm. The twin-trained ACT policy ran on the lab arm without fine-tuning. In a 30-drawing batch (five per class, not the gold-set drawings) every drawing finished, at 0.532 mm mean path RMSE (0.421 to 0.622 mm), 2.7× the twin (Table VI; 2.5 to 3.0× in every class, on different drawings, so compared per class and not paired). Stalls were the main failure mode: 14 of 30 drawings needed at least one nudge to resume (26 in total), always mid-stroke where waypoints are dense; the pen sat about 0.5 mm beside the path and corrected only sideways. Three face portraits drawn with markers (Table VII) finished at 0.497 mm measured RMSE, using an automatic 0.5 s stall release instead of nudges.

Scoring the ink. The arm's reported tip says where the arm believes the pen is, not what reached the paper. We therefore score the sheet: the page is found in an overhead photograph and rectified to millimetres, the ink is thinned to a centre line, and fitted to the plan by a trimmed rigid transform (ink onto plan, so missing ink cannot drag the fit). On portrait A the ink lies 0.43 to 0.48 mm from the plan on average, in line with the reported tip error, and 90% of the plan has ink within 2 mm. Portrait B scores lower (71% covered) because its long hair below the shoulders inked faintly, which the arm's log cannot

TABLE V
PAIRED REAL-ARM COMPARISON, NINE GOLD DRAWINGS, ACT AGAINST THE SHIPPED IK CONTROLLER. IK'S ARM-REPORTED ERROR IS READ AT REST AND IS NOT COMPARABLE. INK COMPLETENESS IS JUDGED BY EYE FROM WRIST-CAMERA PHOTOGRAPHS.

	ACT	IK
Drawings finished	9/9	9/9
Arm-reported path RMSE (mm)	0.458 (0.365-0.518)	not comparable
Time per drawing, mean (s)	**91**	446
Stall releases / manual nudges	22 / 0	0 / 0
Sheets with complete ink	2/9	**8/9**

TABLE VI
ACT ON THE PHYSICAL ARM, 30 ANIMAL DRAWINGS (FIVE PER CLASS). PATH RMSE IN MM.

Class	RMSE mean (range)	Worst point	Time (s)	Nudges / drawing
Bird	0.579 (0.549-0.594)	1.54	76	0.8
Cat	0.488 (0.456-0.539)	1.53	96	1.6
Dog	0.536 (0.484-0.587)	1.47	72	0.0
Fish	0.549 (0.507-0.595)	1.49	102	0.4
Frog	0.480 (0.421-0.569)	1.51	100	0.4
Horse	0.559 (0.442-0.622)	1.54	117	2.0
All	0.532 (0.421-0.622)	1.54	94	0.87

see. The photograph's 0.6 mm pixels bound this measurement; flatbed scans of the paired sheets are still to be scored.

Paired real-arm comparison with IK. We drew nine of the twelve planned gold drawings on the physical arm with ACT and with the shipped IK controller, alternating (two birds, two cats, two dogs, one each of fish, frog and horse; Table V). Here IK is the controller's per-waypoint position command, which stops at every waypoint. Both finished 9/9; ACT drew 4.9× faster (91 against 446 s on average) with 22 automatic stall releases and no manual nudges. The arm-reported path RMSE is informative only for ACT (0.458 mm mean, 0.365 to 0.518 mm): IK is read after it has stopped at each waypoint, so its value is about zero by construction. The like-for-like measure is the sheet, and there IK was better: eight of nine IK drawings were complete in the wrist-camera photographs (two with blobbed small features where waypoints are dense), against two of nine for ACT, whose other seven sheets had gaps. Our run notes attribute these gaps to pen contact rather than path following: the commanded pen height drifted by about ±0.15 mm with position on the sheet, and a fresh round marker tip loses contact 0.1 mm above the paper, whereas the worn, flat-tipped marker of the first two ACT runs gave a clean line. IK settles on the planned height at every waypoint but pools ink where waypoints are dense.

VI. DISCUSSION AND LIMITATIONS

Why ACT beats its own teacher. ACT is trained on IK labels yet outperforms IK because the labels are kinematic: they describe where the arm should go, not IK's gravity-induced sag, and the noise-injected episodes teach correction. Querying a short path window at 60 Hz also replaces IK's stop-and-settle with continuous motion, which is where the jerk and the tremor come from; but continuous motion alone is not enough, since an IK solver chasing a moving target cuts corners and stays jerky.

TABLE VII
FACE PORTRAITS ON THE PHYSICAL ARM WITH ACT AND A MARKER. RMSE, P95 AND MAX ARE FROM THE ARM'S REPORTED TIP AT 10 HZ; CHAMFER AND COVERAGE ARE FROM THE INK IN AN OVERHEAD PHOTOGRAPH. MM.

Run	RMSE	p95	Max	Chamfer	Cov. 2 mm	min
A (48 strokes), used sheet	0.486	0.912	1.306	0.74	90%	8.5
A (48 strokes), clean sheet	0.514	0.966	1.305	0.67	90%	8.3
B (59 strokes), clean sheet	0.490	0.905	1.273	2.13	71%	9.7

Sim-to-real gap. The 2.7× gap is consistent across classes, which points to a systematic cause rather than content. Candidates we observed are a non-level sheet (line weight varies across the page), pen compliance, the arm's own position controller differing from the twin's servo, stalls at densely spaced waypoints, and the real arm's 10 Hz tip logging, which increases the nearest-sample error metric. We have since measured the sheet: touching three corners and the centre with a pencil gave a plane tilted 1.15°, 3.35 mm high across the drawing area, with the centre 0.23 mm below the corner plane, and a later two-contact fit showed the corner-fit slope to be about 1.5× too steep for one sheet stack (0.0200 against 0.0129 mm per mm along x). In the paired study ACT's mean arm-reported error was 0.458 mm, about 2.4× the twin's gold-set mean, but marker and sheet model changed during the study, so we do not claim the gap is explained. Training on stalled states is the next step.

What the pipeline does not preserve. FLUX passes the referee reliably but redraws the subject in a generic pose: in our generator study its median silhouette IoU with the source photo was about 0.2, against 0.78 to 0.91 for OmniGen2's non-blank drawings, and facing direction matched in 65% of cases. The drawings are recognisably the right animal, not a tracing of the photo. Lower diffusion strength or contour conditioning is the obvious next step.

Baseline strength. We compare against the waypoint-gated IK controller the system shipped with and a streaming IK that shares ACT's progress rule. Optimisation-based trackers (model-predictive contouring control [23] or iterative learning control [22]) and the vendor's own Cartesian motion commands on the real arm are stronger baselines we have not yet run.

Limitations. The 48 gold drawings were also used in the development experiments reported in Table III. They therefore informed design decisions and do not constitute a fully untouched test set. Path errors are computed against discrete executed samples, recorded at 10 Hz on the physical arm. This introduces a sampling component that increases with pen speed and the interval between observations. SmolVLA's median query latency of 17.7 ms exceeds the 16.7 ms control period; missed deadlines have not been analysed. Each controller was run once per drawing; ACT was trained with three

seeds; the other policies with one. The 30-drawing real-arm batch used different drawings from the gold set and relied on nudges. The paired real-arm study covers nine of twelve planned gold drawings; marker, sheet-height model and sheet reuse changed between runs, ink completeness was judged by eye from 0.6 mm-per-pixel photographs, and 600 dpi scans are still to be scored, so it is paired but not fully controlled. The path metric is geometric; ink scoring from photographs covers only three portraits so far, and perceived quality (a user study) is future work. The IK baseline is tolerance-gated and its timeouts could be reduced by tuning. Real-arm portraits take 8 to 10 min, against 43 s to 2 min for systems built for exhibition speed [9]–[11]; our runs used the default step scale and stall releases, not a speed-tuned setting. The hybrid supervisor and the camera policies were evaluated on 12 drawings in the interactive runner only.

VII. Conclusion

WildTrace turns photographs into robot pen drawings end to end. A compact ACT-based path follower trained only on kinematic IK labels draws held-out content more accurately, more reliably and more smoothly than the IK solver that labelled it, maintains comparable error across the tested paper placements, and transfers to a physical xArm 7 at about half a millimetre RMSE, with manual interventions to resolve stalls. On nine paired real-arm drawings it is 4.9 times faster than IK, but IK currently inks more completely. A streaming IK with the same progress rule gains almost nothing, so the accuracy cannot be attributed solely to removing the settle gate. A fine-tuned VLA re-planned every tick is slightly more accurate again in a single training seed but jerkier and slower to query, and a simple supervisor repairs injected gaps in the ink bookkeeping for either policy. Closing the remaining sim-to-real gap is our next goal.

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